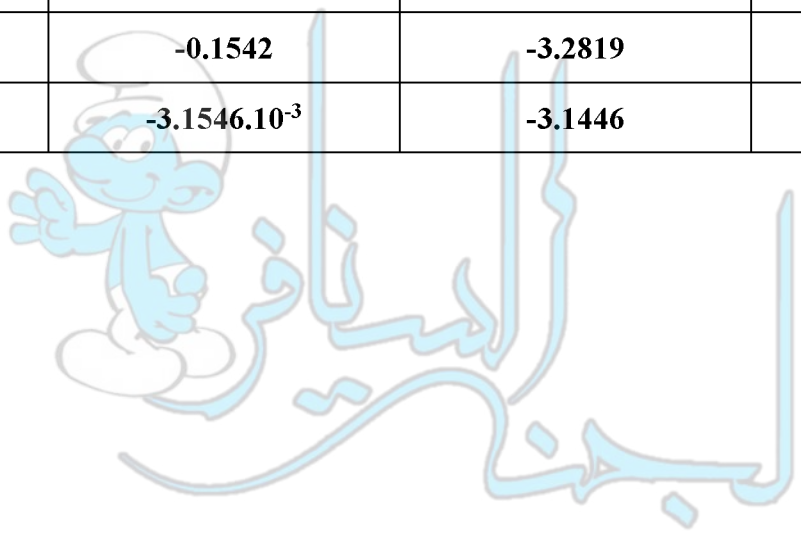
Final Answers

All the computations must be rounded to exactly four decimal places.

Q1- Find the root of the function: $f(x) = 2xcos(x) - sin(x) + 1$

Use the **Newton Raphson Method**. Perform **three iterations** to find the root, assume $x_0 = 1$ **(15 marks)**

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	
0	1	1.2391	-1.1426	2.0844	5 marks
1	2.0844	-1.9192	-4.1223	1.6188	5 marks
2	1.6188	-0.1542	-3.2819	1.5718	3marks
3	1.5718	$-3.1546.10^{-3}$	-3.1446	1.5708	2 marks



Final Answers

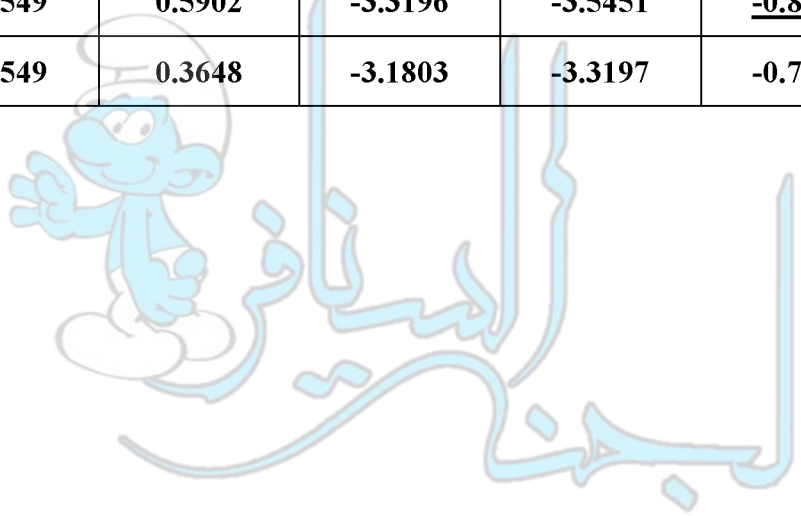
All the computations must be rounded to exactly four decimal places.

Q2- Use the **Golden Section Search** to find the **minimum** of the function:

$f(x) = 5e^x + cos(x)$. Assume that $x_l = -4.5$, $x_U = -2$. Perform **4 iterations.**

(15 marks)

n	x_l	x_U	d	x_1	x_2	$f(x_1)$	$f(x_2)$	
1	-4.5	-2	1.5451	-2.9549	-3.5451	-0.7222	**-0.7754**	5 marks
2	-4.5	-2.9549	0.9549	-3.5451	-3.9098	**-0.7754**	-0.6189	5 marks
3	-3.9098	-2.9549	0.5902	-3.3196	-3.5451	**-0.8034**	-0.7754	3marks
4	-3.5451	-2.9549	0.3648	-3.1803	-3.3197	-0.7914	**-0.8034**	2 marks



Final Answers

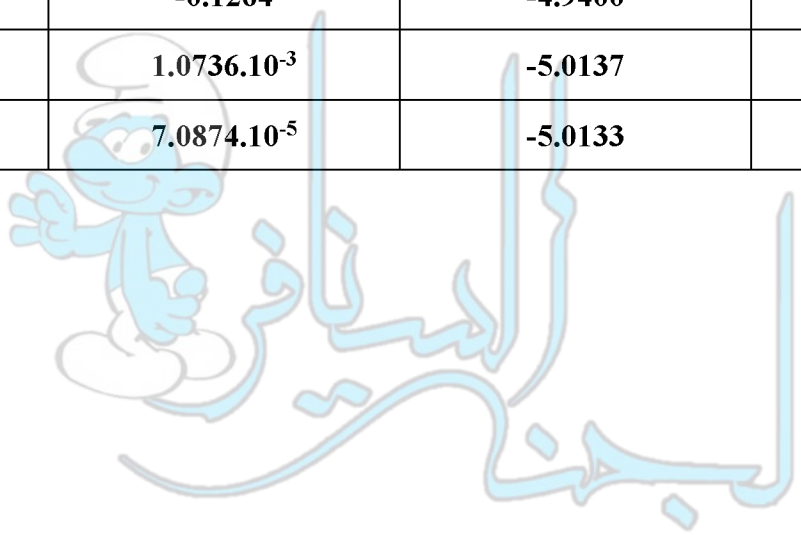
All the computations must be rounded to exactly four decimal places.

Q1- Find the root of the function: $f(x) = 2cos(x^2) - sin(x^2) + 1$

Use the **Newton Raphson Method**. Perform **three iterations** to find the root, assume $x_0 = 1$

(15 marks)

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	
0	1	1.2391	-4.4465	1.2787	5 marks
1	1.2787	-0.1264	-4.9400	1.2531	5 marks
2	1.2531	$1.0736.10^{-3}$	-5.0137	1.2533	3marks
3	1.2533	$7.0874.10^{-5}$	-5.0133	1.2533	2 marks



Final Answers

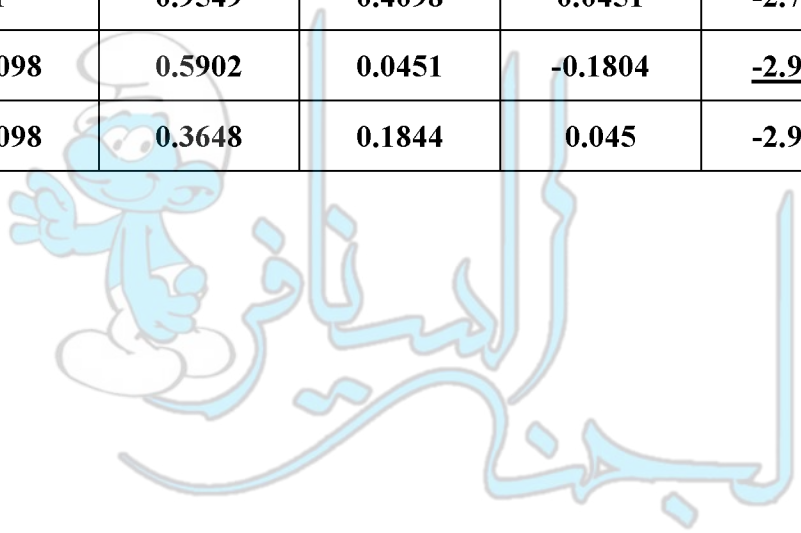
All the computations must be rounded to exactly four decimal places.

Q2- Use the **Golden Section Search** to find the **minimum** of the function:

(15 marks)

$f(x) = 0.2x^3 - 3cos(x)$. Assume that $x_l = -1.5$, $x_U = 1$. Perform **4 iterations.**

n	x_l	x_U	d	x_1	x_2	$f(x_1)$	$f(x_2)$	
1	-1.5	1	1.5451	0.0451	-0.5451	-2.9969	-2.5977	5 marks
2	-0.5451	1	0.9549	0.4098	0.0451	-2.7378	-2.9969	5 marks
3	-0.5451	0.4098	0.5902	0.0451	-0.1804	-2.9969	-2.9525	3marks
4	-0.1804	0.4098	0.3648	0.1844	0.045	-2.9479	-2.9969	2 marks



Final Answers

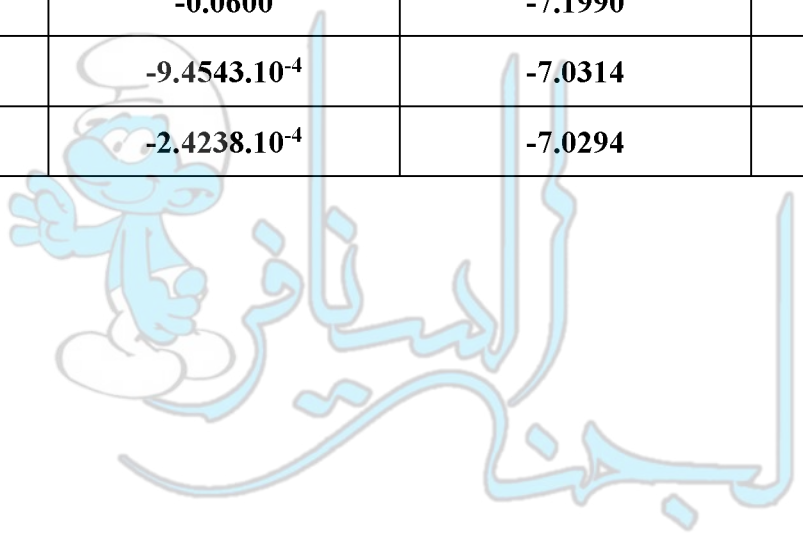
All the computations must be rounded to exactly four decimal places.

Q1- Find the root of the function: $f(x) = 2cos(x^2) - e^{x^2} + 1$

Use the **Newton Raphson Method**. Perform **three iterations** to find the root, assume $x_0 = 1$

(15 marks)

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	
0	1	-0.6377	-8.8024	0.9276	5 marks
1	0.9276	-0.0600	-7.1990	0.9193	5 marks
2	0.9193	$-9.4543.10^{-4}$	-7.0314	0.9192	3marks
3	0.9192	$-2.4238.10^{-4}$	-7.0294	0.9192	2 marks



Final Answers

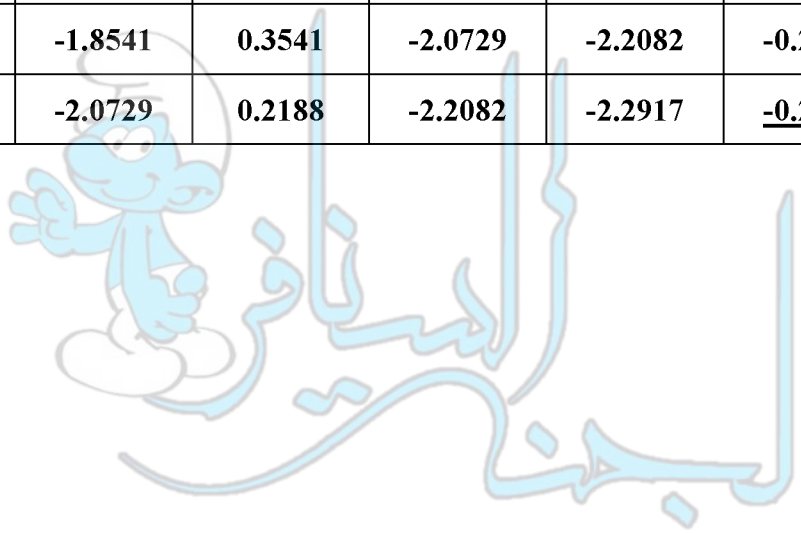
All the computations must be rounded to exactly four decimal places.

Q2- Use the **Golden Section Search** to find the **minimum** of the function:

$f(x) = 5e^x + sin(x)$. Assume that $x_l = -3$, $x_U = -1.5$. Perform **4 iterations.**

(15 marks)

n	x_l	x_U	d	x_1	x_2	$f(x_1)$	$f(x_2)$	
1	-3	-1.5	0.9270	-2.0730	-2.4270	-0.2475	-0.2138	5 marks
2	-2.4270	-1.5	0.5729	-1.8541	-2.0729	-0.1772	-0.2475	5 marks
3	-2.4270	-1.8541	0.3541	-2.0729	-2.2082	-0.2475	-0.2542	3marks
4	-2.4270	-2.0729	0.2188	-2.2082	-2.2917	-0.2542	-0.2457	2 marks



Final Answers

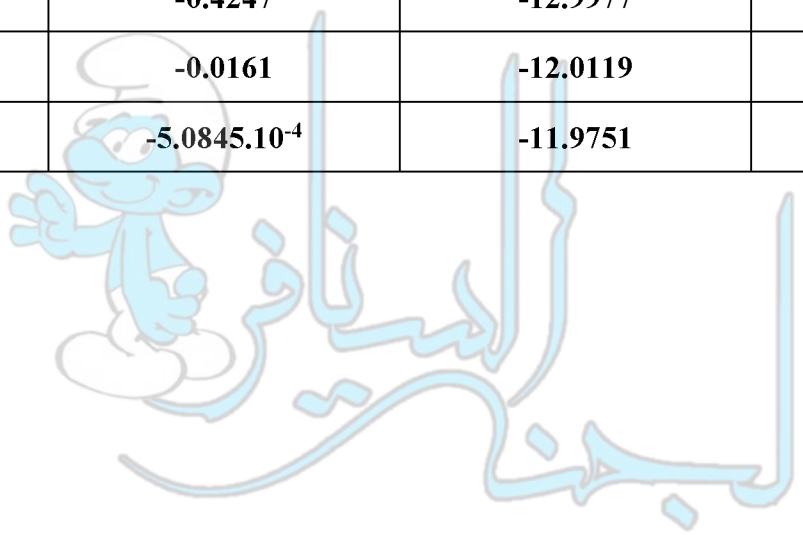
All the computations must be rounded to exactly four decimal places.

Q1- Find the root of the function: $f(x) = 2e^{cos(x)} - e^{x^2} + 1$

Use the **Newton Raphson Method**. Perform **three iterations** to find the root, assume $x_0 = 1$

(15 marks)

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	
0	1	1.7148	-8.3254	1.2060	5 marks
1	1.2060	-0.4247	-12.9977	1.1733	5 marks
2	1.1733	-0.0161	-12.0119	1.1720	3marks
3	1.1720	$-5.0845.10^{-4}$	-11.9751	1.1720	2 marks



Final Answers

All the computations must be rounded to exactly four decimal places.

Q2- Use the **Golden Section Search** to find the **minimum** of the function:

(15 marks)

$f(x) = 0.2x^3 - 3sin(x)$. Assume that $x_l = 0$, $x_U = 2$. Perform **4 iterations.**

n	x_l	x_U	d	x_1	x_2	$f(x_1)$	$f(x_2)$	
1	0	2	1.2361	1.2361	0.7639	-2.4558	-1.9861	5 marks
2	0.7639	2	0.7639	1.5278	1.2361	-2.2839	-2.4558	5 marks
3	0.7639	1.5278	0.4721	1.2360	1.0557	-2.4558	-2.3754	3marks
4	1.0557	1.5278	0.2918	1.3475	1.2360	-2.4362	-2.4558	2 marks

